

ShadowSparrow: An Ambient Display for Information Visualization and Notification

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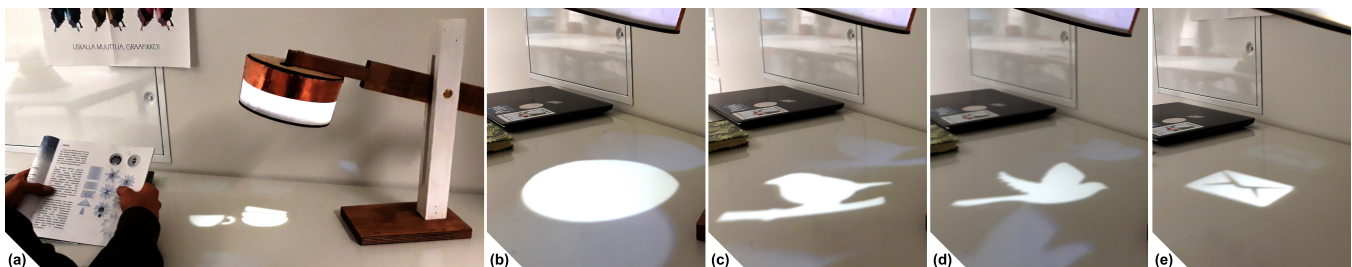


Figure 1: ShadowSparrow ambient display. (a) Dishwasher icon is projected while the user is reading a book; (b) Empty circle state; (c) Standing bird state; (d) Flying bird state, (e) E-mail icon state.

ABSTRACT

We present the design and prototype of ShadowSparrow, an ambient display utilizing lights & shadows. The concept uses illustrative bird images and other relevant visuals (e.g., dishwasher and e-mail icons) cast on the surroundings for ambient information visualizations and notifications. Our work creates an alternative channel to convey non-critical information in an unobtrusive and aesthetic way.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Ambient Displays, shadows, pervasive displays, unobtrusive design, ambient notification, information visualization.

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1 INTRODUCTION

Two different human needs, to be in a calm environment and yet be informed, are typically in conflict with each other [12]. One solution to this problem is to take advantage of peripheral vision. Ambient displays, which are the antithesis of conventional display screens [13], allow users to easily and quickly move their focus of attention between different displays, e.g. based on their location in the physical environment [10, 11]. In contrast to conventional graphical user interfaces (GUI), ambient displays provide an alternative channel for human-computer interaction that acts within peripheral vision [8, 13]. Nature provides inspiration for subtle, expressive ambient displays [7] that awaken our senses [13].

In this paper, we present an aesthetic ambient display concept, *ShadowSparrow*, that carries information through functional lights and shadows, Figure 1. It enables an unobtrusive and calm background channel by utilising playful illustrations of a bird's actions and posture, as well as other visuals (e.g., dinnerware and e-mail icons), to carry less critical information. The ShadowSparrow allows a seamless transition between different activities. The concept gives insight into the use of functional shadows as decorative elements embedded in the physical environment.

Lights & shadows are a promising design direction, as they are by their very nature always present, yet unobtrusive, part of our everyday surroundings. Although ambient displays have been much researched in the last two decades, the use of shadows as part of human-computer interaction, has received little attention. Shadow play and puppet theatre, one of the earliest examples, have been around for centuries [2, 9]. Shadows as part of ambient displays

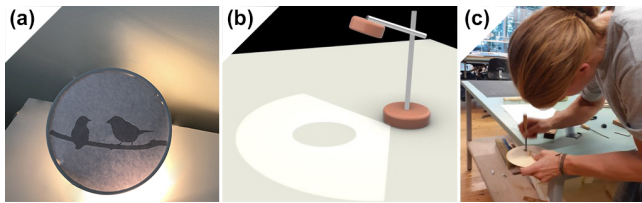


Figure 2: Design process. (a) Non-functional prototype; (b) 3D design; (c) Constructing the ShadowSparrow.

have been demonstrated in relatively few concepts. In the ambient-ROOM [5], the use of light, shadows and sound were investigated. Colley et al. [1] presented a shadow manipulation display prototype where non-critical information was embedded into the shadow of a house plant. Although ambient displays are designed to convey information, they can also appear as aesthetic decorative objects integrated into daily life. Häkkinen et al. [3] presented a candle shadow display that delivers subtle and playful cues using emoji-style images. In ShadowLamp [6], Jensen et al. used electrochromic displays as a mask to project imagery from a children’s book on a ceiling for ambient communication.

2 CONCEPT, PROTOTYPE & SCENARIOS

2.1 The Design

ShadowSparrow ambient display (Figure 1) aims to utilize nature elements that create a calming and comforting atmosphere [4] and uses the lights and shadows as the heart of the display. The concept builds on the previous findings introduced by Colley et al. [1]. When updated sensory information is received, the light silhouette changes, representing the information content. In creating the concept, we drew inspiration from the actions and gestures of a bird, and aimed to reflect this in a dynamic visual dialog.

The design was done in iterative manner (Figure 2), by first creating a non-functional prototype of a shadow display. The low fidelity prototype of the ShadowSparrow was built from two metal hoops covered with a textured fabric. As the light source, we used a desk lamp to cast the cutout bird images. The concept was then iterated further, and the form factor changed to a display where the light is projected onto the surroundings, in order to create more of an impression of a shadow and less of a screen (Figure 2-b). At this stage, we also experimented with other relevant visuals, such as a dinnerware or an e-mail icon, to explore direct symbolic associations with inputs from the Internet of Things (IoT) use-cases.

2.2 Implementation & Scenarios

The ShadowSparrow utilizes a working principle where the light is cast through a disc with cutouts (Figure 3). By rotating the disc, the display controls the light on the surface based on the shape of the hole below the light source. To enable this, the cylindrical lamp head incorporates a servomotor connected to laser-cut disc with cutouts, a smartphone flashlight module as the light source for casting shadows, an RGB LED strip for aesthetic lighting, a micro-processor (Adafruit HUZZAH32) to control the lights and rotate the disc, as well as a Li-Po (3.7 V) battery to power the setup. The smartphone flashlight module is selected since they provide a focused

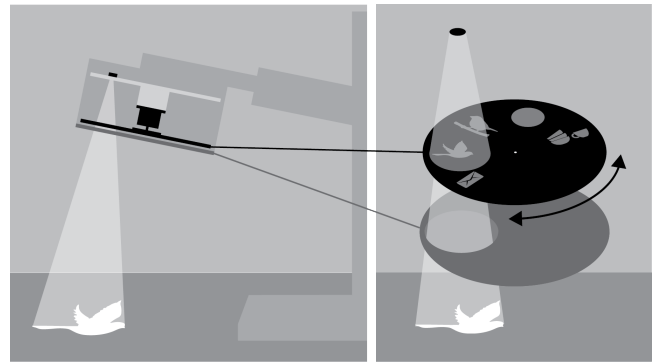


Figure 3: Working principle of the ShadowSparrow.

light, aiding clear shadows. Also, Adafruit HUZZAH32 is used as this board allows wireless connection, over Wi-Fi or Bluetooth, to various devices (e.g., smartphones, smart home appliances).

This setup enabled us to explore usage scenarios for ambient notifications and visualizations. During the explorations, the rotation of the disc was controlled by using the Serial Bluetooth Terminal application on an Android smartphone. In the *first scenario*, we focus on displaying an ambient visualization of a timer with the states of a bird: The user connects to the ShadowSparrow and sets a timer for studying. At this stage, the display casts empty circle (Figure 1-b). Halfway through the end, the disc rotates and the visual turns into a standing bird (Figure 1-c). Finally, when the time is over, the display shows a flying bird (Figure 1-d). In the *second scenario*, the ShadowSparrow is connected to smart home system and casts an dinnerware icon when the dishwasher is done (Figure 1-a). The *third scenario* showcases a subtle notification via an e-mail icon when an important e-mail arrives.

3 DISCUSSION AND CONCLUSION

In this paper, we have presented the design and prototype of the ShadowSparrow ambient display, based on the visuals created by light and shadow, as a novel ambient display concept to carry non-critical information. We aimed to design an aesthetically pleasing human-computer interaction experience where shadows are used to express information. Although, our preliminary prototype of the concept is not yet ready for real-world deployment, it demonstrates the potential of the approach. In the next iterations of the prototype, we plan to employ it in real world settings for an in-the-wild study.

Creating shadows as a physical demo creates challenges e.g. in the visual presentation quality, and a high-quality outcome requires careful mechanical design. Visual factors such as the sharpness of the shadow and distortion in the dimensions may affect the user experience (UX). On the other hand, this is the case with natural, real-world shadows. Thus, a projected shadow may be more forgiving for UX than digital representations of artificial shadows. This would be an interesting direction for future studies. To conclude, we believe that our concept prototype provides an inspiring vision for the future of ambient communication systems.

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